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# Design of Binary Decision Diagram (BDD) Optical Adder

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***Abstract*— In this paper a new design technique for 4-bit carry look ahead optical adder based on binary decision diagram (BDD) has been proposed. This technique can be used for the design of higher order bits. Performance analysis is carried out for longest and shortest path lengths in terms of power attenuation. In this paper, the threshold value of output power to interpret as logic 1 or logic 0 are also determined.**

***Keywords*—** *BDD, Optical Implementation, Adders, Paths, Output Power*

### *Introduction*

All-optical signal processing concepts and technologies have evolved remarkably in the past few years, primarily due to the significant advancement of semiconductor-based, all-optical, switching devices. A novel technique of performing optical logic operations is proposed by Changxi Yang et al. [1]. This technique is based on beam coupling in photo refractive self-pumped phase conjugators. The system is capable of performing real time and spatially parallel operations. An implementation of All-Optical Full-Adder is proposed by Ghelfi et al. [2]. This photonic digital processing fundamental block is obtained by cascading a single integrable logic gate exploiting XGM and XPM-based polarization rotation in Semiconductor optical Amplifier (SOA).

It is required to design optical processing circuits using minimum number of optical gates to achieve the power efficiency after considering the complexity of all-optical switching devices. D.Tsiokos et al. [3] have proposed a data processor to perform operations like binary addition, counting, data recognition and encoding based on binary half-adder as the basic building block. A simple all-optical module that uses two ultra-fast non-linear interferometer (UNI) gates and operates at 10 G bits to perform a total of four Boolean logic operations is proposed by Kehayas et al. [4].

An all-optical half adder using semiconductor optical amplifier based devices is suggested and demonstrated at 10 Gbit/s by Jae Hun Kim et al. [5]. Boolean AB and AB out of two semiconductor optical amplifiers have been obtained and combined to achieve an all-optical XOR gate, which is logic SUM. Also, an all-optical XPM wavelength converter has been utilized to obtain an all-optical AND gate, which is logic CARRY. Logic SUM and CARRY are simultaneously realized.

A common method to implement digital logic function is by using logic gates. Logic gates requires transistors. Adders have been built with transistors. [6-8]. Realization using Complementary Metal Oxide Transistors (CMOS) are extremely popular. They require many transistors and inherently slow. Pass Transistor Logic (PTL) offers a good area/power delay trade-off alternative to static CMOS circuits. At present, Very Large Scale Integrated circuits (VLSI) deploy transistors in one form or the other. Optical circuits are low power consumption and high speed alternatives for the current information processing based on transistor circuits. An important advantage of BDD in terms of fabrication is that it is amenable for implementation in widely different forms. Thus it can be implemented using transistors or optical principles or field techniques and so on.

In principle, BDDs can be used to represent any Boolean function. BDDs are canonical and efficient way to represent and manipulate Boolean functions and have been successfully used in numerous CAD applications [9-12]. The popularity of BDDs in the analytical studies of digital circuits encouraged researches to fabricate devices using this concept. Hence it is possible to build a BDD device for any digital application. It is generally known that BDDs are compact, but no compact realizations are readily available for any combinational circuits viz. adders.

A desire to realise all-optical computing using digital optical elements begin several years ago. All-optical processors are becoming demanding in the high capacity core networks. Various architectures have been proposed in the field of optical/optoelectronic computing. The BDD is also an optics friendly architecture to do optical logic computing. A scalar signal like an optical pulse is launched into the root of a circuit

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as a messenger. Asahi et al. [13] have demonstrated the single-electron logic circuits based on the BDD in which a single electron is used as the messenger. They have also proposed the designs of several basic logic circuits and predicted that using photons instead of a single-electron as the signal could also achieve the same functionality. An optical computing logic based on binary decision diagram is demonstrated by Shiyun Lin et al [14].

The present paper discusses the design of BDD based 4-bit carry look ahead optical adder. The designed optical adder is studied and analyzed in detail in terms of power loss and threshold optical power to represents the logic 1 and logic 0. This technique can be used to implement any bit size adder.

## 2. Preliminaries

This section introduce the basics of Adders and Binary Decision Diagram

### 2.1 Carry look-ahead Adder

All digital computers contain logic circuits to implement arithmetic operations in a particular number system. The elementary arithmetic modules include binary addition, binary subtraction, binary multiplication and binary division. Two's complement number system is usually used, as it simplifies the representation of both positive and negative values, and the implementation of addition and subtraction circuits.

The Full adder sum and Carry expressions are given in eqn (1) and (2). The adders constructed cascading Full adders (ripple adders) face the delay problems.

$$C_{i+1} = x_i y_i + (x_i + y_i) c_i \quad (1)$$

$$S_i = (x_i \oplus y_i) \oplus C_i \quad (2)$$

To eliminate the ripple effect , the carry output $C_{i+1}$ at any given stage is expressed in terms of the operand variables themselves, i.e. $x_i$ and $y_i$, and the initial input carry $C_0$ ($C_0$ = 0), so that the ripple effect is eliminated and the overall speed of the adder increases. Divide the full adders into groups and employ bypass logic to speed up the carry propagation. The four bit carry look ahead adder is shown in Figure 1.

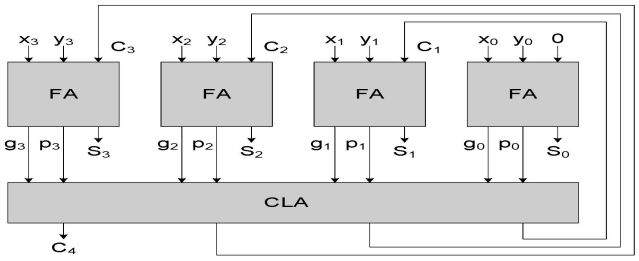



Figure 1: Four Bit Carry Look Ahead Adder

$$g_i = x_i y_i \quad (3)$$

$$p_i = x_i + y_i \quad (4)$$

$g_i$: carry-generate term for stage *i*
$p_i$ : carry-propagate term for stage *i*

The carry output at *i*th stage is

$$C_{i+1} = g_i + p_i c_i \quad (5)$$

$$C_1 = g_0 + p_0 c_0 \quad (6)$$

$$C_2 = g_1 + p_1 c_1 = g_1 + p_1 g_0 + p_1 p_0 c_0 \quad (7)$$

$$C_3 = g_2 + p_2 c_2 = g_2 + p_2 g_1 + p_2 p_1 g_0 + p_2 p_1 p_0 c_0 \quad (8)$$

$$C_{i+1} = g_i + p_i g_{i-1} + p_i p_{i-1} g_{i-2} + \ldots + p_i p_{i-1} \ldots p_1 g_0 + p_i p_{i-1} \ldots p_0 c_0 \quad (9)$$

### 2.2 Binary Decision Diagrams

For an efficient data structure, representation and manipulation of Boolean expressions are essential. Various methods have been developed to perform efficient manipulation of Boolean expressions. Truth tables, cube sets are the traditional methods used. Truth tables generally need $2^n$ bit of memory to represent an n-input function. The memory requirement grows exponentially with respect to the input variables. This also leads to exponential computation time. Hence the truth tables are not suitable for Boolean expressions with large number of input variables.

Cube sets or sum-of-products (SOP), PLA forms, covers, or two level logics are the special form of Boolean expressions with AND-OR two level structure. The representation is certainly better than truth tables but redundant cubes may appear in logic operations and they have to be reduced in order to check equivalency. Hence the limitation is that the reduction process is time consuming. Other limitations are NOT operation cannot be performed easily and the size of the cube sets of parity function becomes exponential. So the traditional methods are impractical for large Boolean functions. It has been necessary to develop an efficient data structure that can handle large Boolean functions.

Akers was the first researcher who has developed the basic concept of BDDs in 1978. BDDs are mainly graph representation of Boolean functions. The efficient manipulations of BDDs were developed by Bryant [6]. Ordered Binary Decision Diagrams (OBDDs) referred to the BDDs of fixed variable ordering. Reduced Ordered Binary Decision Diagrams (ROBDDs) referred to the BDDs after applying the reduction rules to a fixed variable ordering BDDs.

BDD is a directed acyclic graph (DAG). The two sink nodes, labelled 0 and 1, represent the Boolean functions 0 and 1. Each non-sink node is labelled with a Boolean variable v and $\bar{v}$ has two out edges labelled 1 (then) and 0 (else). Each non-sink node represents the Boolean function corresponding

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to its 1 edge if v = 1 or the Boolean function corresponding to its 0 edge if v = 0. An ordered binary decision diagram (OBDD) is a BDD with the constraint that the input variables are ordered, and every source to sink path in the OBDD visits the input variables in an ascending order. A reduced ordered binary decision diagrams (ROBDD) is an OBDD where each node represents a distinct logic function. The size of the ROBDD is influenced by variable ordering. An important property of ROBDDs is that they are a canonical representation of Boolean functions.

The fundamental property which makes BDDs as a state-of-art data structure is Canonicity i.e., The ROBDD of a Boolean function for a predefined ordering of variables is unique. In addition, BDDs are the only data structure that has graph algorithms for all important Boolean operations.

An arbitrary Boolean function graph can be simplified to a reduced graph which represents the same function. It can be achieved by using the simple reduction mechanism. The reduction rules include elimination and merging rules.

BDDs are the only data structures for the representation of switching functions that have a deterministic polynomial algorithm for important operations such as evaluation, minimization, an equivalence test, and a satisfiability test. BDD is able to solve many problems through symbolic analysis.

The Shannon expansion is very important for Boolean algebra because it paved the way for Binary decision diagrams. It develops the concept that Boolean functions can be reduced by way of the identity. Each node represents Shannon expansion along a variable [6]:

$$F = xFx \; + x'Fx' \tag{10}$$

where $F$ is any function, and $Fx$ and $Fx'$ are positive and negative Shannon cofactors of $F$, respectively.

## 3. Design and Implementation

The source file is generated for the 4-bit carry look ahead adder. By using the source file as an input for the BDD generation package and decide on the optimal BDD size by checking the variable ordering methods. The BDD package ***CUDD-2.4.2*** [15] has been used for drawing the BDD with different variable ordering technique. The optimal BDD size obtained by using symmetric sifting to convergence ordering technique is shown in Figure 2. The 4-bit BDD adder is implemented using ***OptiSystem s***oftware for optical modelling. It is constructed from bottom to the top.

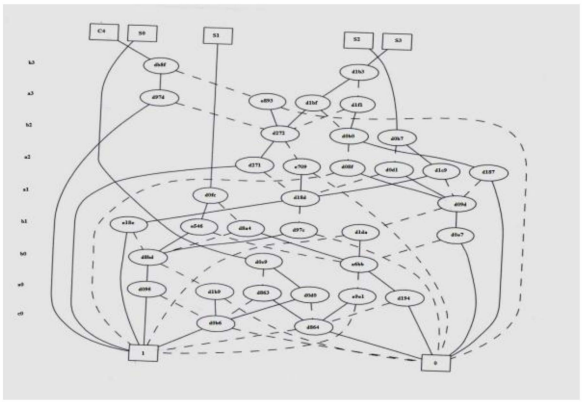


**Figure 2: 4 Bit BDD Adder for Symmetrical sifting Convergence ordering Technique**

### *3.1 Top level design of BDD*

Figure 3 shows the top level design of the 4-bit BDD. In the top level design the BDD nodes are shown as a block with inputs and Outputs. It consists of 2 sets of 4-bit input signal ($a_0$, $a_1$, $a_2$, $a_3$, $b_0$, $b_1$, $b_2$, $b_3$) and carry bit ($c_0$) on the top left hand side of the block. The laser source with power 5dBm, frequency of 1550nm is fed to the logical 1 input acting as a light source. In terms of Boolean representation, presence of light represents logic 1 whereas the absence of light represents logic 0. The light source is placed at the bottom of the BDD. The optical power meter is used to measure the power of the light signal that reaches at the output of the network. The result of the binary addition is known by measuring the power of the signal output of the network namely $S_0$, $S_1$, $S_2$, $S_3$ and $C_4$. The optical spectrum analyzer shows the peak power at wavelength 1550 nm and the noise of the signal. The Fork 1×N is used since it copies the input signal into a user-defined number of output signals. It allows the user to duplicate component output ports. It is used to copy the control signals in order to be connected to multiple switches.

The optical implementation of nodes of the BDD is explained in detail with proper illustration. Each node in BDD is represented by using the Optical Y Select switch as shown in Figure 4(a). When the control signal is 0, optical signal at input 1 is passed through to the output port and for a control signal 1 the input 2 is passed to the output port as shown in Figure 4(b). This kind of behavior is desired in the BDD application in order to guide the light source to the designated output. Referring to the node of the BDD, the input 2 signals of the switch is arranged in such a way that the line that carries value 1 is connected to the input 1 and value 0 to the input 2. The arrangement of the inputs is maintained throughout the whole BDD.

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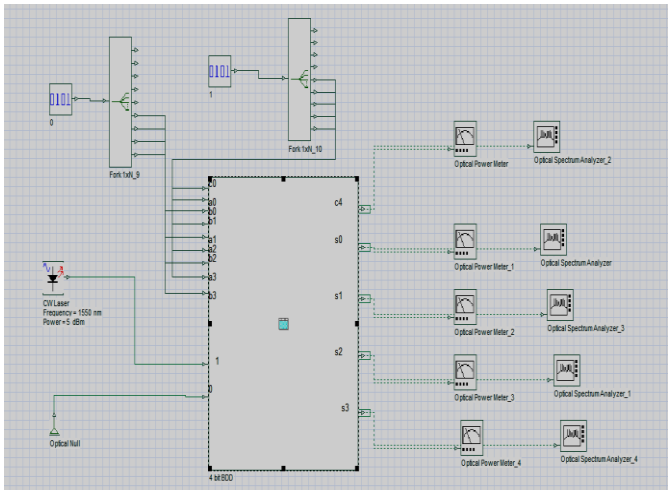


**Figure 3: Top level design of the BDD**

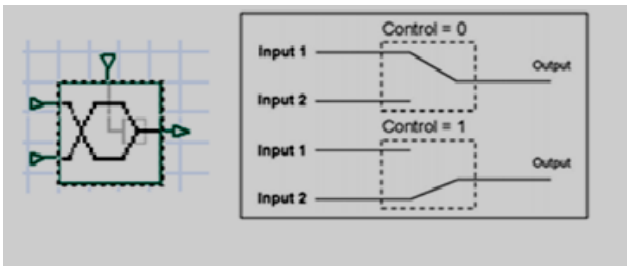



**Figure 4: (a) Optical Y-Select (b) Switch Behaviour**

The control signal is binary in nature which can be supplied by the user defined bit sequence generator. The bit sequence generator supplies binary number as the input to the control signal of the Optical Switch to decide which input signal port to be passed to the output. In the BDD network, the light source might need to be split into multiple paths so that it can be connected to the next consecutive stage of the nodes. For this purpose, the power splitter is used to split the input signal into user-defined number of output signals.

The optical switches must be connected in a specific notation throughout the BDD. The 4 bit BDD adder ($c_0$ and $a_0$) is shown in Figure 5. The BDD solid line represents logic 1 and dashed line represents logic 0. The control signal of $c_0$ and $a_0$ determine through which input line signal propagate to the output. To implement the nodes into the optical switch, the solid and dashed lines must be connected to Input 2 and Input 1 of the optical switch.

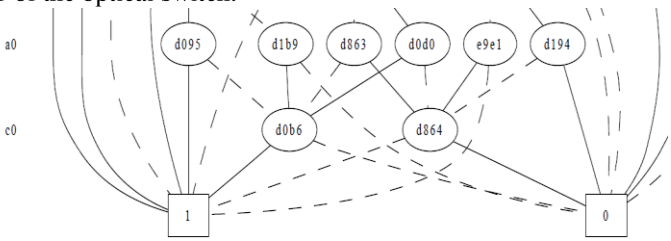



**Figure 5: BDD nodes for $c_0$ and $a_0$**

The optical implementation of the BDD nodes for $c_0$ and $a_0$ is shown in Figure 6. The power splitter splits the light source to the inputs of optical switches. The control signals from the bit sequence generator are connected to the fork 1×N in order to be distributed to the multiple optical switches for the variables $c_0$ and $a_0$. Following the BDD, for $a_0$ there are two non-sink nodes and they are replaced by two optical switches and control signal is generated by bit sequence generator and connected through Fork 1x2. The control signal is duplicated and connected to 2 switches. The output of the nodes d096 and d864, have four output lines each. So power splitters are used to split the output light into four and connected to the next level switches. For the second level, there are five non-sink nodes and they are replaced by five optical switches and are controlled by the signal generated by the bit sequence generator and connected through the Fork1x5. The control signal is duplicated and connected to 5 switches.

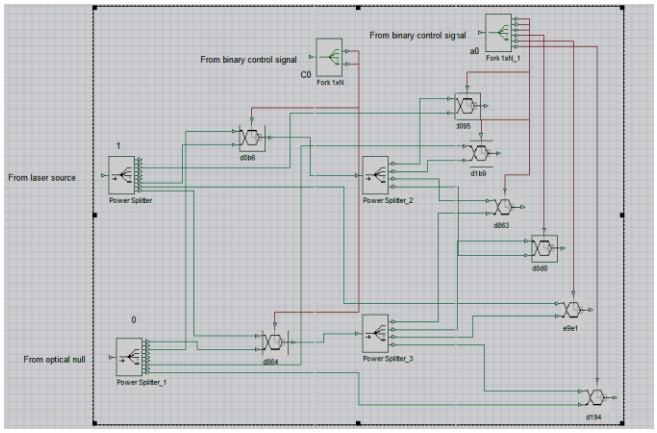



**Figure 6: Optical implementation of BDD for $c_0$ and $a_0$**

Figure 7 shows the optical switch implementation of BDD nodes of 4 bit adder. The light source is guided by the optical switches network depending on the control signal of the switches. The control signals represent the desired 4 bit binary addition that we would like to perform.

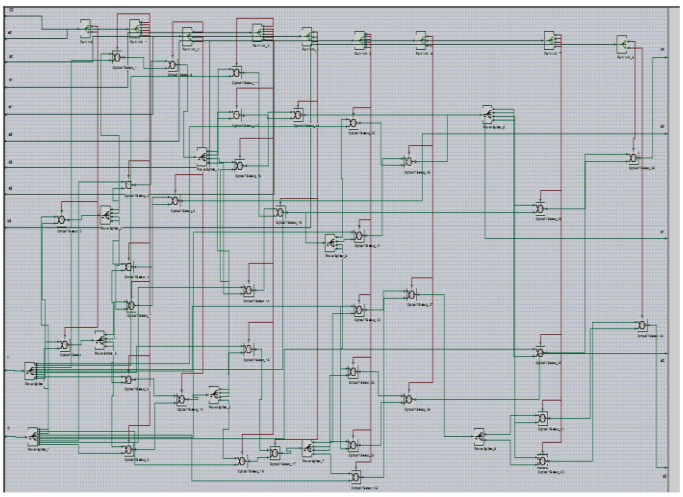


**Figure 7: Optical switches network of the 4 bit BDD Adder**

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## 4. Results and Discussions

### *4.1 Longest and shortest path*

The simulation is carried out by supplying the binary control signal to the BDD adder. These signals guide the light source to the output of the network which are observed at $S_0$, $S_1$, $S_2$, $S_3$ and $C_4$. The optical network of the BDD is analyzed by considering the shortest and longest path. To determine the threshold power of the light to be considered as logic 1 or logic 0, a series of control inputs are applied into the package.

By analyzing the BDD network, the longest path for any output means that the light source needs to travel through all of the 9 nodes in BDD. For example, the path taken by light to travel to $S_3$ requires control signal is shown in Table 1.

It is observed that the power output of $S_3$ is -31.913 dBm. This value of power must equal to logical 1 since the expected binary output at S3 is 1. On the other hand, it is observed that the output power of $C_4$, is -61.22 dBm which is equivalent to logic 0. The optical spectrum analyzer outputs are given in Figure 8 which shows the spectrum for Logic 1 and Logic 0.

To determine the range for logic 0 and 1, another set of control signals, as given in Table 2, is simulated. By analyzing the BDD, the shortest path can be determined at output $C_4$ when $a_3$ and $b_3$ is 1. The threshold power required to represent logic 1 and logic 0 for shortest path is given in Table 2. The spectrum analyzer results for shortest path are shown in Figure 9.

Another set of simulation is carried out for random values and the result is given in Table 3. At the output of BDD, the optical spectrum analyzer gives the value of the peak power at frequency 1550nm. From the observations of Table 1 to 3, it is concluded that the range of power to be considered as logical 1 lies between -5.9 dBm to -32 dBm whereas for logical 0, the output power must less then -60dBm.

### 4.2 Performance analysis

The longest path taken is the light source to propagate through all of the nodes for output $S_3$. The light intensity is attenuated as it propagates through the optical switches network depending on the number of switches and power splitter it has to go through. The total power loss for the longest path is 36.916 dBm. The shortest path for the light source to travel in the BDD is the output $C_4$ when $a_3$ and $b_3$ are logical 1. The total power loss for the shortest path is 10.942 dBm. The comparison of power loss for the shortest and longest path is shown in Figure 10. Figure 11 shows the power loss with respect to number of splitters and switches. From the graph, it is clear that the power loss increases with increase in the number of splitters and switches.

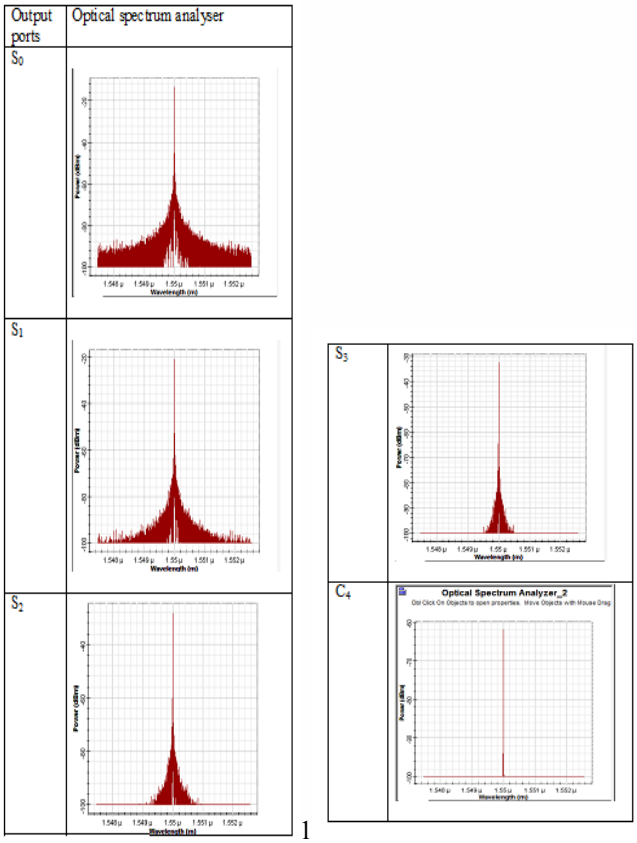



**Figure 8: Optical spectrum analyzer for longest path of $S_3$**

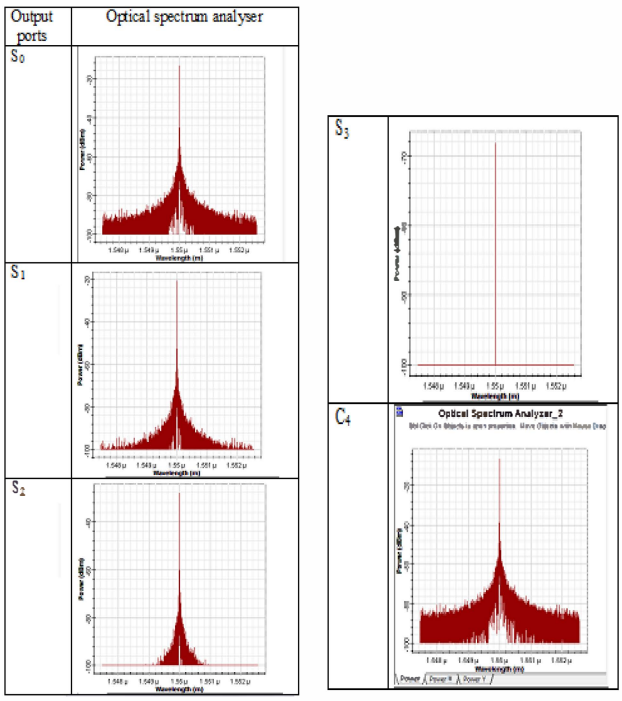



**Figure 9: Optical spectrum analyzer for shortest path of $C_4$**

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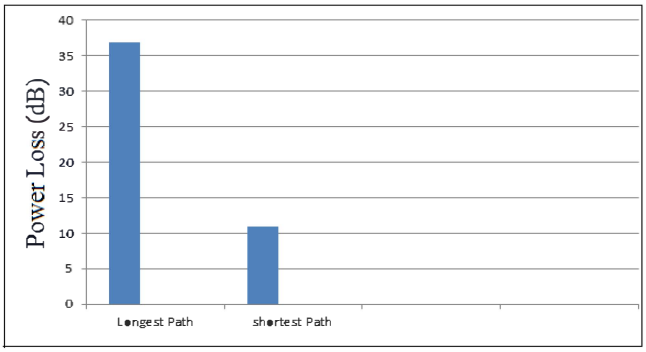



**Figure 10: Power loss of the 4Bit BDD optical adder shortest and Longest Paths**

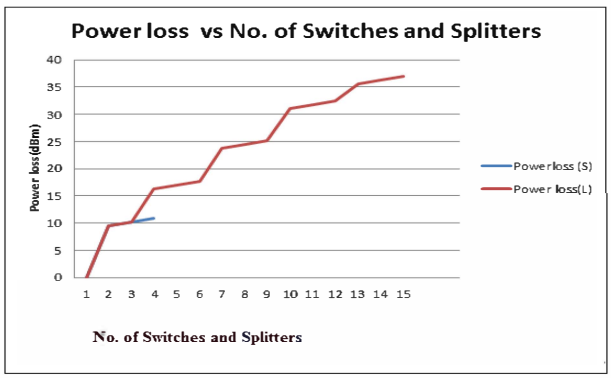



**Figure 11: Power loss for the longest and shortest path of the 4 bit BDD optical adder**

## 5. Conclusion

The BDD based 4 Bit optical adder has been proposed and its performance is analyzed for shortest and longest path lengths. From simulation it is observed that optical power in the range of -5.9 dBm to -32 dBm is interpreted as logic 1 whereas for logic 0, the output power must be less than -60dBm. Optical adders of higher number of bits (16, 32, 64 etc) can be easily implemented by combining two or more 4-bit adder as required. It can be used in the design of Optical Arithmetic Logic Unit (ALU).

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**Table 1: Longest path for output $S_3$**

| Control Signal | Result of binary addition | Result of simulation | | | | |
|---|---|---|---|---|---|---|
| Symbols | Binary value | Symbols | Expected binary value | Symbols | Output power (dBm) | Equivalent binary value |
| $a_0$ | 1 | $S_0$ | 1 | $S_0$ | -12.663 | 1 |
| $a_1$ | 0 | $S_1$ | 1 | $S_1$ | -20.084 | 1 |
| $a_2$ | 1 | $S_2$ | 1 | $S_2$ | -27.504 | 1 |
| $a_3$ | 1 | $S_3$ | 1 | $S_3$ | -31.913 | 1 |
| $b_0$ | 0 | $C_4$ | 0 | $C_4$ | -61.220 | 0 |
| $b_1$ | 1 | | | | | |
| $b_2$ | 0 | | | | | |
| $b_3$ | 0 | | | | | |
| $c_0$ | 0 | | | | | |

**Table 2: Shortest path for $C_4$**

| Control Signal | Result of binary addition | Result of simulation | | | | |
|---|---|---|---|---|---|---|
| Symbols | Binary value | Symbols | Expected binary value | Symbols | Output power (dBm) | Equivalent binary value |
| $a_0$ | 1 | $S_0$ | 1 | $S_0$ | -12.663 | 1 |
| $a_1$ | 0 | $S_1$ | 1 | $S_1$ | -20.084 | 1 |
| $a_2$ | 1 | $S_2$ | 1 | $S_2$ | -27.504 | 1 |
| $a_3$ | 1 | $S_3$ | 0 | $S_3$ | -67.574 | 0 |
| $b_0$ | 0 | $C_4$ | 1 | $C_4$ | -5.942 | 1 |
| $b_1$ | 1 | | | | | |
| $b_2$ | 0 | | | | | |
| $b_3$ | 1 | | | | | |
| $c_0$ | 0 | | | | | |

**Table 3: Random control input signal for the BDD**

| Control Signal | Result of binary addition | Result of simulation | | | | |
|---|---|---|---|---|---|---|
| Symbols | Binary value | Symbols | Expected binary value | Symbols | Output power (dBm) | Equivalent binary value |
| $a_0$ | 1 | $S_0$ | 1 | $S_0$ | -12.663 | 1 |
| $a_1$ | 1 | $S_1$ | 0 | $S_1$ | -65.678 | 0 |
| $a_2$ | 1 | $S_2$ | 0 | $S_2$ | -65.782 | 0 |
| $a_3$ | 1 | $S_3$ | 0 | $S_3$ | -64.93 | 0 |
| $b_0$ | 0 | $C_4$ | 1 | $C_4$ | -20.783 | 1 |
| $b_1$ | 1 | | | | | |
| $b_2$ | 0 | | | | | |
| $b_3$ | 0 | | | | | |
| $c_0$ | 0 | | | | | |

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